

María Alejandra Schild Zerene

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Studies

Pontificia Universidad Católica de Chile (UC Chile) <i>PhD in Engineering Sciences, specializing in Computer Science</i>	Santiago, Chile 2025-present
Pontificia Universidad Católica de Chile (UC Chile) <i>Master's in Mathematics</i> ○ GPA: 6.6/7.0 ○ Thesis title: <i>Computational Complexity of Counting Shellings of a Simple Graph</i>	Santiago, Chile 2022-2024
Pontificia Universidad Católica de Chile (UC Chile) <i>Bachelor's in Mathematics</i> ○ GPA: 6.82/7.0	Santiago, Chile 2018-2021

Awards and Scholarships

National Agency for Research and Development (ANID) <i>ANID National Doctoral Scholarship</i> Ranked 10th out of 3359 applications	Chile 2025
Chilean Mathematical Society (SOMACHI) <i>Honorable Mention, Jorge Billeke Award</i> Awarded for academic excellence among graduates of the Bachelor's programs in Mathematics and Mathematical Engineering	Chile 2022
National Agency for Research and Development (ANID) <i>ANID National Master's Scholarship</i> Ranked 6th out of 1895 applications	Chile 2022
Pontificia Universidad Católica de Chile (UC Chile) <i>Academic Excellence Award</i>	Santiago, Chile 2021
Faculty of Mathematics, UC Chile <i>Rolando Chuaqui Scholarship</i> Award for academic excellence	Santiago, Chile 2020
International Mathematical Olympiad (IMO) <i>Bronze Medal</i>	Rio de Janeiro, Brazil 2017

Teaching

Escuela Militar del Libertador Bernardo O'Higgins <i>Introduction to Programming</i> for second-year cadets, under an academic agreement with Pontificia Universidad Católica de Chile	Santiago, Chile 1st semester 2026
Faculty of Mathematics, UC Chile <i>Introduction to Geometry</i> for students in the Bachelor in Mathematics program and in the secondary mathematics teacher education program	Santiago, Chile 2nd semester 2024
Faculty of Mathematics, UC Chile <i>Introduction to Calculus</i> for students in the Bachelor in Computer Science program	Santiago, Chile 1st semester 2024
Faculty of Mathematics, UC Chile <i>Instructor in the Mathematical Reasoning Workshop (TRM) program</i>	Santiago, Chile April-November 2022

Teaching Assistantships

Department of Computer Science, UC Chile <i>Topics in Computer Science (course topic: communication complexity)</i> Grading assistant	Santiago, Chile 2nd semester 2025
Department of Computer Science, UC Chile <i>Logic for Computer Science</i> Teaching assistant	Santiago, Chile 1st semester 2025
Faculty of Mathematics, UC Chile <i>Foundations of Mathematics; Introduction to Combinatorics; Calculus I</i> Teaching and grading assistant	Santiago, Chile 2022-2024
Faculty of Mathematics, UC Chile <i>Mathematics Help Center (twice); Mathematics Workshop (twice)</i>	Santiago, Chile 2019-2021

Talks

Computational Complexity Seminar (Department of Computer Science, UC Chile) <i>Search-to-decision reducibility for NP problems (1 session)</i>	Santiago, Chile April 2026
Computational Complexity Seminar (Department of Computer Science, UC Chile) <i>Computational complexity of the Tutte polynomial (2 sessions)</i>	Santiago, Chile September 2025
Computational Complexity Seminar (Faculty of Mathematics, UC Chile) <i>Connections between algebraic combinatorics and the complexity class #P (8 sessions)</i>	Santiago, Chile August-November 2023
Algebraic Combinatorics Seminar (Faculty of Mathematics, UC Chile) <i>Shelling orders and Stanley-Reisner rings (3 sessions)</i>	Santiago, Chile May-June 2023
Algebraic Combinatorics Seminar (Faculty of Mathematics, UC Chile) <i>Deletion-contraction and the Tutte polynomial of matroids (2 sessions)</i>	Santiago, Chile September 2022
Mathematics Club (Faculty of Mathematics, UC Chile) <i>The Kruskal-Katona theorem for simplicial complexes (1 session)</i>	Santiago, Chile June 2022

Conferences and Schools Attended

Schloss Dagstuhl - Leibniz Center for Informatics <i>Knowledge Compilation in Artificial Intelligence, Databases, and Formal Methods</i>	Wadern, Germany May 26-29, 2026
Max Planck Institute for Mathematics in the Sciences <i>Summer School in Algebraic Combinatorics</i>	Leipzig, Germany July 15-19, 2024
Encuentro Colombiano de Combinatoria (ECCO) <i>Bridges between Algebra and Combinatorics</i>	Popayán, Colombia June 17-21, 2024
CIMPA Research School <i>Crossroads of Geometry, Representation Theory and Higher Structures</i>	Puerto Madryn, Argentina March 13-24, 2023

Languages

- Spanish: Native
- English: Fluent
- German: Intermediate

Other Activities

Pan-American Girls' Mathematical Olympiad (PAGMO)
Coordinator and member of the Problem Selection Committee

Online
2021 and 2022